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RAMAIAH Institute of Technology

(Autonomous Institute, Affiliated to VTU)
(Approved by AICTE, New Delhi & Govt. of Karnataka)
Accredited by NBA & NAAC with 'A+' Grade

SEMESTER END / BACKLOG SUBJECT EXAMINATIONS – FEB / MARCH 2025

Program	:	**B.E. – Common to CSE / ISE / AI & ML / AI & DS / CSE (AI & ML) / CSE (CY)**	**Semester**	:	**III**
Course Name	:	**Laplace Transforms and Vector Space**	**Max. Marks**	:	**100**
Course Code	:	**CS / IS / AI / AD / CI / CY31 (2022,2023 Batch)**	**Duration**	:	**3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

1. a) Find $L\left\{e^t\int_0^t \frac{\sin t}{t}dt\right\}$. CO1 (06)

   b) A periodic function of period $2\pi/\omega$ is defined by $f(t)=\begin{cases} E\sin(\omega t) & for\ 0<t<\pi/\omega \\ 0 & for\ \pi/\omega<t<2\pi/\omega \end{cases}$, where E and ω are positive constants, then show that $L\{f(t)\}=\dfrac{E\omega}{(s^2+\omega^2)(1-e^{-\pi s/\omega})}$. CO1 (07)

   c) If $L\{f(t)\}=F(s)$, then prove that $L\{t^n f(t)\}=(-1)^n\dfrac{d^n[F(s)]}{ds^n}$, where 'n' is a positive integer. CO1 (07)

2. a) Evaluate $\int_0^\infty t\,e^{-3t}\sin t\,dt$. CO1 (06)

   b) If $\mathrm{L}(\mathrm{f(t)})=F(s)$, then prove that $\mathrm{L}\left(\dfrac{\mathrm{f(t)}}{\mathrm{t}}\right)=\int_s^\infty F(s)\,ds$. Also, evaluate $L\left(\dfrac{\sin 2t}{t}\right)$. CO1 (07)

   c) Obtain the Laplace transform of (i) $\left(\sqrt{t}+\dfrac{1}{\sqrt{t}}\right)^3$ (ii) $L\{te^{-4t}\sin 3t\}$. CO1 (07)

UNIT - II

3. a) Express in terms of Heaviside unit step function and find the Laplace transform of $f(t)=\begin{cases}1 & 0<t\le 1\\ t & 1<t\le 2\\ t^2 & t\ge 2\end{cases}$. CO2 (06)

   b) Find the inverse Laplace transform of $\dfrac{5s+3}{(s-1)(s^2+2s+5)}$. CO2 (07)

   c) Solve $y''+4y'+4y=e^{-t}$ with $y(0)=0\ and\ y'(0)=0$ using Laplace Transform. CO2 (07)

4. a) Find $L^{-1}\left(\log\left(\frac{s^2+1}{s(s+1)}\right)\right)$. CO2 (06)

b) Using Convolution Theorem Evaluate $L^{-1}\left[\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right]$. CO2 (07)

c) Using Laplace transform, solve the simultaneous linear differential equation $\frac{dx}{dt}-2y=\cos 2t,\quad \frac{dy}{dt}+2x=\sin 2t$ given that $x(0)=1, y(0)=0$. CO2 (07)

UNIT - III

5. a) Define basis and dimension of a vector space V and determine whether the set $\left\{ \begin{bmatrix}6 & 0\\0 & 0\end{bmatrix}, \begin{bmatrix}2 & 3\\1 & 3\end{bmatrix}, \begin{bmatrix}2 & 3\\4 & 1\end{bmatrix}, \begin{bmatrix}2 & 3\\4 & 5\end{bmatrix} \right\}$is basis over the vector space M_{22} of set of all 2x2 matrices. CO3 (06)

b) Verify Rank nullity theorem for the transformation T(x, y) = (3x, x-y, y) of $R^2 \rightarrow R^3$. CO3 (07)

c) Prove that the transformation T: $R^2 \rightarrow R^2$ given by T (x, y) = (2x, x-y) is linear. Find the images of the elements (1, 2) and (-1, 4) under this transformation. CO3 (07)

6. a) Determine the matrix that defines a reflection in the y-axis followed by rotation through an angle $\pi/6$ and then by the dilation of factor 3/2 . Find the image of the point $\begin{bmatrix}2\\1\end{bmatrix}$ under this transformation. CO3 (06)

b) Define linearly independent vectors with an example. Prove that the set of vectors {(1, -2, 3), (-2, 4, 1), (-4, 8, 9)} is linearly dependent in R^3. Express one vector in each set as a linear combination of the other vectors. CO3 (07)

c) Find the transition matrix P from the basis B to the standard basis B' of R^2. Use this matrix to find the coordinate vectors u, v, and w relative to B', where B={(4, 1), (3, -1)} and B' = {(1, 0), (0, 1)}; $u_B = \begin{bmatrix}3\\0\end{bmatrix}$ $v_B = \begin{bmatrix}2\\1\end{bmatrix}$ $w_B = \begin{bmatrix}-2\\1\end{bmatrix}$. CO3 (07)

UNIT- IV

7. a) Find the complete solution of $A=\begin{pmatrix}1 & 0 & 2 & 3\\1 & 3 & 2 & 0\\2 & 0 & 4 & 9\end{pmatrix}$ and b=$\begin{pmatrix}2\\5\\10\end{pmatrix}$. CO4 (10)

b) Find the dimension and basis for four fundamental subspaces of CO4 (10)

$$A=\begin{bmatrix}1 & 2 & 0 & 1\\0 & 1 & 1 & 0\\1 & 2 & 0 & 1\end{bmatrix}.$$

8. a) Find QR factorization of CO4 (10)

$$A=\begin{bmatrix}1 & 0 & 0\\1 & 1 & 0\\1 & 1 & 1\\1 & 1 & 1\end{bmatrix}$$

b) Find the least-squares solution of $Ax=b$ CO4 (10)

Where A=$\begin{pmatrix}1 & 1 & 0 & 0\\1 & 1 & 0 & 0\\1 & 0 & 1 & 0\\1 & 0 & 1 & 0\\1 & 0 & 0 & 1\\1 & 0 & 0 & 1\end{pmatrix}$ and b=$\begin{pmatrix}-3\\-1\\0\\2\\5\\1\end{pmatrix}$

UNIT - V

9. a) Diagonalize the matrix $A=\begin{bmatrix}1 & 4\\ 1 & -2\end{bmatrix}$ and hence find A^5. CO5 (10)

b) i) Express the quadratic form $Q(x)=x_1^{\ 2}-8x_1x_2-5x_2^{\ 2}$ as quadratic forms with no-cross product terms . CO5 (10)

ii) Determine Unitary matrix *P* that diagonalize the matrix $A=\begin{bmatrix}4 & 2+2i\\ 2-2i & 6\end{bmatrix}$.

10. a) Find an SVD of the $A=\begin{bmatrix}1 & -1\\ -2 & 2\\ 2 & -2\end{bmatrix}$. CO5 (10)

b) i) Check whether the matrix $A=\begin{bmatrix}3 & -1 & 1\\ -1 & 5 & -1\\ 1 & -1 & 3\end{bmatrix}$ is positive definite. CO5 (10)

ii) Determine whether the matrix $A=\begin{bmatrix}3 & 7-4i & -2+5i\\ 7+4i & -2 & 3+i\\ -2-5i & 3-i & 4\end{bmatrix}$ are Hermitian.

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